Categorical Physics Learning Paradigm

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Abstract

We propose a framework that represents the process of problem solving in physics in terms of functorial language. We represent physical phenomena as the category **Phys**, and establish '**represent**' functors between **Phys** and the category of problem solving, denoted **Think**. Natural transformations between **Think** categories allow for a framework that captures misconceptions.

1 The Category Phys

Let $O \in ob(\mathbf{Phys})$ if O is a physical observation. Let $\tau \in Hom(O_1, O_2)$ if τ is an evolution that takes the observers physical observation from O_1 to O_2.

Identity morphisms represent no discernible change in the physical observation. Composition of morphisms represents a sequence of evolutions.

2 The Category Think

Let $P \in ob(\mathbf{Think})$ if P is an understanding the observer has. Then $T \in Hom(P_1, P_2)$ is a thought process that carries the observer from understanding P_1 to understanding P_2. Composition is a sequence of thought processes.

The identity morphism is represents the observer repeating this understanding.

3 Represent Functor

Let $R : \mathbf{Phys} \rightarrow \mathbf{Think}$ be the functor that maps the observers physical observations and evolutions thereof into the category **Think**.

4 Misconceptions as Natural Transformations

The natural transformation $\alpha : F_1 \Rightarrow F_2$ represents the transition of thought processes between paradigms of thought. If F_1 is a functorial misconception, whereas F_2 represents a non-misconception thought paradigm, then $\alpha : F_1 \Rightarrow F_2$ gives the correction of thought processes.

5 Conclusion

Do to the finickiness of epistemological discovery, the overmathematization of the learning process would do more harm than good. However, this categorical framework provides a general enough allowance for epistemological argumentation that it remains unrestrictive. It also gives a rigorous enough formalization via category theory that it will aid learners in modeling learning processes in future research. The current limitation of this framework is the lack of metacognition. An enrichment would allow for metacognition via processes about processes; modeling the **Represent** functors faithfully remains an issue though. Metacognition would provide a generalized framework that allows for adaptive learning paradigms in the student.